



A

Project Report On

“DIGITAL VIBE”

Towards partial fulfillment of the requirement in

Master of Science In Information Technology & Computer Application

Submitted by

ACADEMIC YEAR: 2025-2026

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Project Guide

Prof. Mitalben Vaishnav



Submitted To:

**Shree Matrumandir Institute of Information Technology &
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This is to certify that Chavda Suraj Mukeshbhai
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project Report on Digital Vibe as
partial fulfillment of M.Sc.(IT & CA) Semester - 2 Programme,
Saurashtra University.

We wish them all the best for their bright future.

Project Guide

Examiner

Shree MatruMandir Institute of Information Technology & Management

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ACKNOWLEDGEMENT

First and foremost, we would like to express our sincere gratitude to our respected guide, **Prof. Mitalben Vaishnav**, for her valuable guidance, continuous support, and motivation throughout the completion of this project. We are truly thankful to her for providing us with this wonderful opportunity to work on the project “**Digital Vibe**”.

This project has given us hands-on exposure to various concepts related to **E-commerce systems**, website development, and database management. Working on this project has enhanced our technical knowledge and improved our practical understanding of web technologies.

We are also thankful to all our faculty members of **Shree Matrumandir Institute of Information Technology & Management** for their constant encouragement and for sharing their knowledge, which helped us at every stage of this project.

Finally, we would like to thank our classmates, friends, and everyone who directly or indirectly supported us during the development and testing phase of this project.

PREFACE

The project “**Digital Vibe**” is developed to provide users with a seamless and modern online shopping experience for a wide range of electronic products. The platform focuses on delivering the latest gadgets, home appliances, and tech accessories that cater to different user needs, preferences, and budgets.

At Digital Vibe, we believe that online shopping should be fast, secure, and user-friendly. Therefore, the website is designed using modern web technologies such as **React.js** for a dynamic and responsive user interface, ensuring smooth navigation and an interactive browsing experience. Customers can easily explore products, compare specifications, and make purchases from anywhere at their convenience.

All products available on the platform are well-organized into categories, making it easier for users to find what they are looking for. The system also ensures product authenticity, detailed descriptions, and updated pricing to build user trust and satisfaction. Our aim is to combine innovation with reliability, giving customers both quality products and a secure transaction environment.

The platform includes essential e-commerce features such as a shopping cart, secure checkout process, and multiple payment options, supported by a robust backend developed using **Node.js** and a **MongoDB** database for efficient data management.

We also emphasize continuous improvement by considering user feedback and integrating new features based on customer needs and market trends. This project represents our practical understanding of full-stack web development and modern e-commerce architecture.

Thank you for exploring our project. We hope that **Digital Vibe** successfully demonstrates an efficient, scalable, and user-friendly online electronics shopping platform.

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1. Project Summary

- 1.1 Project Profile.**
- 1.2 Project Purpose.**
- 1.3 Project Development Model.**
- 1.4 Project Development Scheduling.**
- 1.5 Project scope.**
- 1.6 Technology Used.**
- 1.7 Feasibility Study.**

1.1 Project Profile:

Project Title : Digital Vibe

Institution : Rajkot

Front End : HTML, CSS, JavaScript, Bootstrap, React js

Back End : Node js, Mongo DB

Team Size : 2 Members

Project Guide : Prof. MITALBEN VAISHNAV

Submitted By : Chavda Suraj Mukeshbhai

And Khatra Nensi Tejasbhai

Submitted To : Shree Matrumandir Institute of Information Technology & Management

1.2 Project definition - purpose

The purpose of the **Digital Vibe** project is to provide users with a reliable and user-friendly online platform for purchasing electronic products. It allows customers to easily browse, select, and buy gadgets and accessories from anywhere.

The main goal is to develop a secure and efficient e-commerce system using modern technologies like **React.js**, ensuring smooth navigation, clear product information, and a hassle-free shopping experience.

Additionally, the platform is designed to build customer trust by offering accurate product information, multiple payment options, and reliable transaction handling through a robust backend using **Node.js** and **MongoDB**.

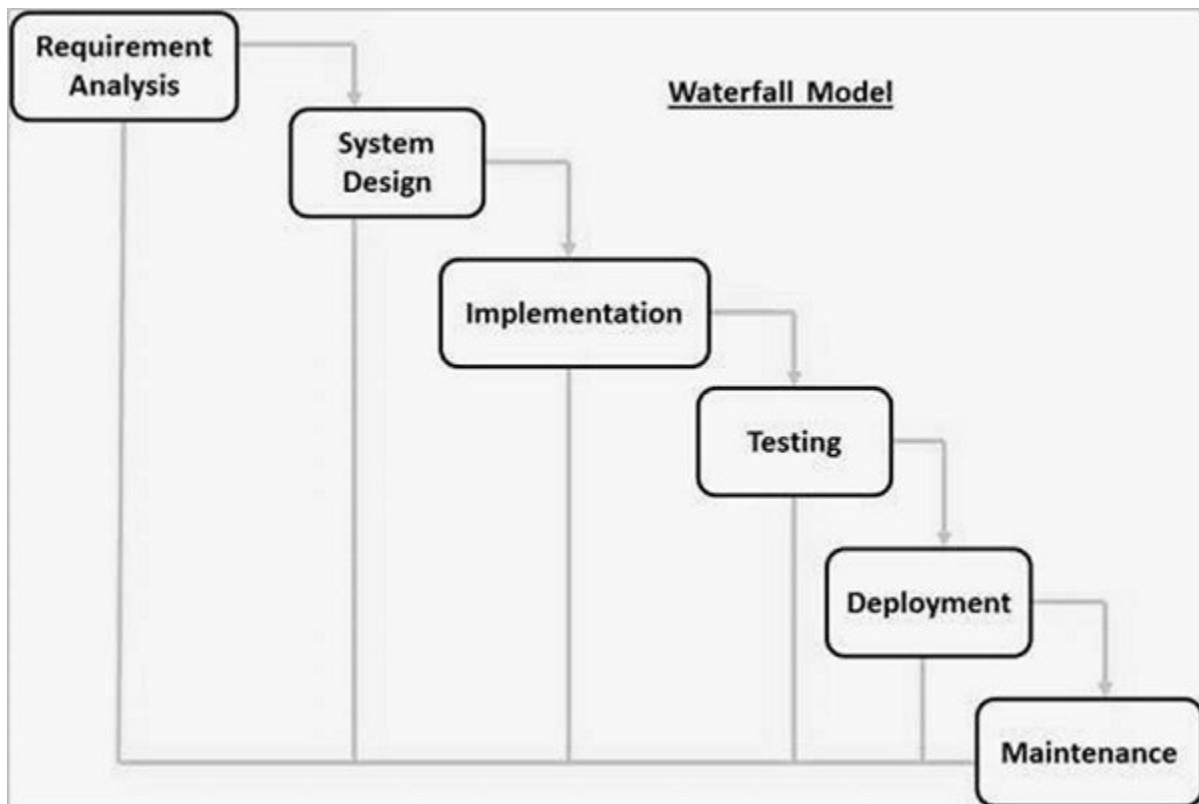
Overall, Digital Vibe aims to create a simple, convenient, and trustworthy platform that enhances customer satisfaction in online electronics shopping.

1.3 Project Development Model

A software development model is a simplified description of the software process that shows how a project is planned and executed. The choice of a suitable model depends on the type of project, its requirements, tools, and the expected outcomes.

For **Digital Vibe**, a systematic development model was followed to ensure smooth and organized progress at every stage. The project is treated as a problem-solving process that passes through four main stages:

- **Status Quo:** Represents the current situation, identifying the need for an online platform for purchasing electronic products easily.
- **Problem Definition:** Defines the problem of limited accessibility to electronic products in offline stores and the need for a convenient and reliable online shopping system.
- **Technical Development:** Develops the solution using technologies like HTML, CSS, JavaScript, Bootstrap, **React.js**, **Node.js**, and **MongoDB**.
- **Solution Integration:** Delivers the final working e-commerce website with all features such as product browsing, shopping cart, secure checkout, and complete system functionality to meet user requirements.



1.4 Project Development Scheduling:

Week Month	Week 1	Week 2	Week3	Week 4
1 st Month	Requirement Gathering	Initial Design	Database Setup	Frontend Development
2 nd Month	Backend Development	Integration of Modules	Testing (Phase 1)	Bug Fixes & Improvements
3 rd Month	User Testing	Final Debugging	Project Deployment	Post- Deployment Support

1.5 Project scope.

The **Digital Vibe** project aims to develop a reliable and modern online platform for purchasing electronic products, offering a smooth and efficient shopping experience for users. The website provides a user-friendly interface where customers can easily browse and search for various gadgets, home appliances, and accessories, with options to filter by price, brand, or category. Registered users can create and manage their accounts to save favorite products, view order history, and receive personalized recommendations based on their preferences.

The platform includes secure and multiple payment options to ensure safe online transactions. Customers can track their orders, receive updates on order status, and contact support for assistance or queries. The scope also includes an effective inventory management system that allows the admin to manage product listings, stock availability, and sales records efficiently.

In addition, the project integrates features like customer reviews, ratings, and discount offers to increase user engagement and trust. Promotional deals and special offers are included to enhance customer satisfaction and encourage repeat purchases. The system also focuses on a smooth delivery process to ensure timely and accurate shipment of products.

Overall, the scope of **Digital Vibe** is to create a complete, secure, and scalable e-commerce solution that provides a convenient and trustworthy online shopping experience for electronic products.

1.6 Technology Used.

What is React.js?

React.js is a popular open-source JavaScript library used for building modern and dynamic user interfaces, especially for single-page applications. It allows developers to create reusable UI components, making the development process faster and more efficient. React uses a virtual DOM to improve performance by updating only the necessary parts of a web page instead of reloading the entire page. It works seamlessly with HTML, CSS, and JavaScript to create interactive and responsive front-end applications. React.js is widely used for building scalable and high-performance user interfaces.

What is Node.js?

Node.js is a powerful server-side JavaScript runtime environment that allows developers to build fast and scalable web applications. It uses an event-driven, non-blocking architecture, making it efficient for handling multiple requests at the same time. Node.js enables developers to use JavaScript for both front-end and back-end development, simplifying the development process. It is widely used for building APIs, handling server logic, and managing data flow between the client and the database.

What is HTML?

HTML (Hypertext Markup Language) is the basic structure of any website. It defines the layout of content such as text, images, links, and forms on a web page. Web browsers read HTML documents and display them to users as visual web pages. HTML is easy to learn and forms the foundation upon which other technologies like CSS, JavaScript, and Bootstrap work together to make websites dynamic and attractive.

What is CSS?

CSS (Cascading Style Sheets) is used to design and format web pages created with HTML. It defines how the elements should look — including colors, fonts, layout, and spacing. By separating content from design, CSS makes it easier to manage and update website styling. A single CSS file can be used across multiple pages to ensure consistent design and improve website speed. CSS also helps make web pages responsive so they look good on all devices.

What is JAVASCRIPT?

JavaScript is a lightweight programming language that adds interactivity and dynamic features to web pages. It runs directly in the web browser, allowing actions like form validation, animations, and live content updates without reloading the page. JavaScript makes websites more engaging and user-friendly. It is supported by all modern browsers and works closely with HTML and CSS to create powerful front-end experiences.

What is MongoDB ?

MongoDB is a NoSQL, open-source database used to store and manage large amounts of unstructured data. Unlike traditional databases, it stores data in flexible, JSON-like documents, making it easy to scale and modify. MongoDB provides high performance, scalability, and flexibility, making it suitable for modern web applications. It works efficiently with Node.js to handle backend operations such as data storage, user management, and application logic.

1.7 Feasibility Study.

A feasibility study is an essential step carried out before starting the **Digital Vibe** project to determine whether it is practical, achievable, and worth developing. It helps in evaluating if the project can be successfully implemented using the available time, budget, and resources. This study includes analyzing key aspects such as technical feasibility (availability of technologies like React.js, Node.js, and MongoDB), economic feasibility (cost-effectiveness of development and maintenance), legal feasibility (compliance with online transaction and data security regulations), operational feasibility (ease of use for customers and administrators), and schedule feasibility (completion within the planned timeline).

By carefully evaluating these factors, the feasibility study helps identify possible challenges at an early stage, reduce risks, and support effective decision-making. It also provides a clear understanding of the project's strengths, limitations, and growth opportunities. Overall, the main objective of conducting this feasibility study is to ensure that the **Digital Vibe** project is technically reliable, economically viable, and capable of delivering a successful and efficient e-commerce solution.

2. System requirement & specification.

2.1 User Characteristics.

2.2 User access flow.

2.3 Current system.

2.4 Proposed system & advantage.

2.4.1 Proposed system.

2.4.2 Advantage of proposed system.

2.5 Hardware & Software requirement.

2.5.1 Hardware requirement.

2.5.2 Software requirement.

2.1 User characteristics

In the **Digital Vibe** project, users can perform a variety of essential functions that make their shopping experience smooth and convenient:

1. **Browse and Search for Products:** Users can explore different categories such as smartphones, laptops, accessories, and home appliances. They can also use the search bar and filters to quickly find specific products.
2. **View Product Details:** Each product has a detailed page displaying information like price, specifications, features, images, and customer reviews to help users make informed decisions.
3. **Add to Cart and Wishlist:** Users can add selected products to their shopping cart for purchase or save them to their wishlist for future reference.
4. **Place Orders:** Customers can proceed to checkout, review selected items, choose payment methods, enter delivery details, and confirm their orders easily.
5. **Manage Orders:** Users can view their order history, track current orders, and manage their purchases efficiently.
6. **Feedback and Support:** Users can rate products, provide feedback, and contact customer support for any queries or issues.

This system ensures a simple, user-friendly, and efficient online shopping experience, guiding users from product browsing to purchase and post-sale support.

2.2 User access Flow

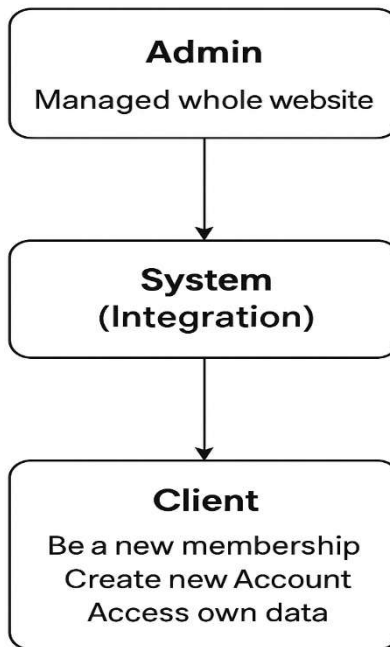


Figure 2.2 – User Access Flow

2.3 Proposed system & advantages

Proposed System

The proposed system is a modern, role-based e-commerce platform developed using **React.js** for the front-end and **Node.js** with **MongoDB** for the back-end. It includes both an admin panel and a user-facing website, designed to ensure efficiency, security, and scalability.

The admin panel allows authorized users to manage key modules such as products, categories, users, and orders through a well-structured interface.

On the other hand, customers can browse electronic products, add items to their cart or wishlist, and complete purchases easily. The system focuses on providing a secure, responsive, and user-friendly experience across different devices.

Advantages of the Proposed System

1. Scalability:

The system is designed to handle a growing number of users, products, and transactions. With the use of modern technologies, it can easily scale as the platform expands.

2. Security & Role Management:

Role-based access control ensures that only authorized admins can manage system data. Secure authentication and data handling help protect user information and transactions.

3. User-Friendly Experience:

The platform provides a clean and interactive interface. Users can easily browse products, manage their cart, and complete purchases with features like wishlist, order tracking, and secure checkout.

4. Efficient Management:

All operations such as product management, user handling, and order processing are centralized, reducing manual effort and improving system efficiency.

2.4 Software& hardware requirement

2.4.1 Hardware Requirement:

To run and test the **Digital Vibe** project smoothly, the following minimum hardware configuration is required:

- Processor: Intel Core i3 or higher
 - RAM: Minimum 4 GB (8 GB recommended)
 - Hard Disk: 500 GB or more
 - Monitor: Standard Display (15" or higher)
 - Keyboard & Mouse: Standard Input Devices
-

2.4.2 Software Requirement:

The project is developed using modern web technologies with a full-stack JavaScript approach. Below are the required software tools and versions for proper setup and execution:

- Operating System: Windows 10 / 11 or Ubuntu Linux
- Front-End Technologies: HTML, CSS, JavaScript, Bootstrap, **React.js**
- Back-End Technologies: **Node.js** (latest LTS version)
- Database: **MongoDB**
- Server & API Tools: Express.js (for backend development)
- Package Manager: npm (Node Package Manager)
- Code Editor: Visual Studio Code / Sublime Text
- Browser: Google Chrome / Mozilla Firefox (latest version)

3. Design System

3.1 Context Diagram.

3.2 Dataflow Diagram.

3.3 ER diagram.

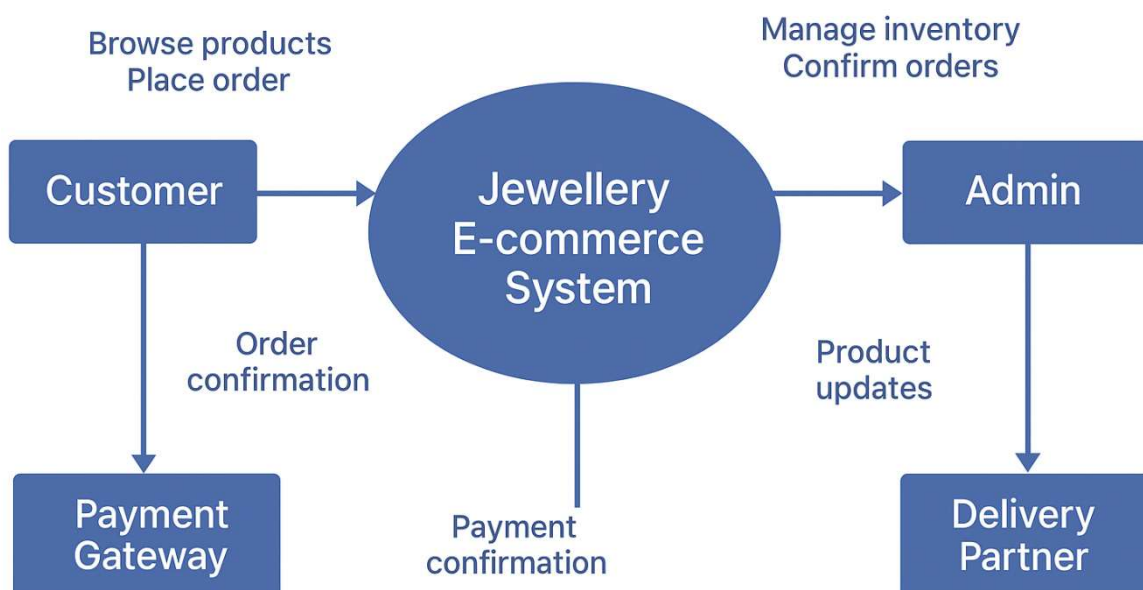
3.4 Use Cases

3.5 Flow Chart

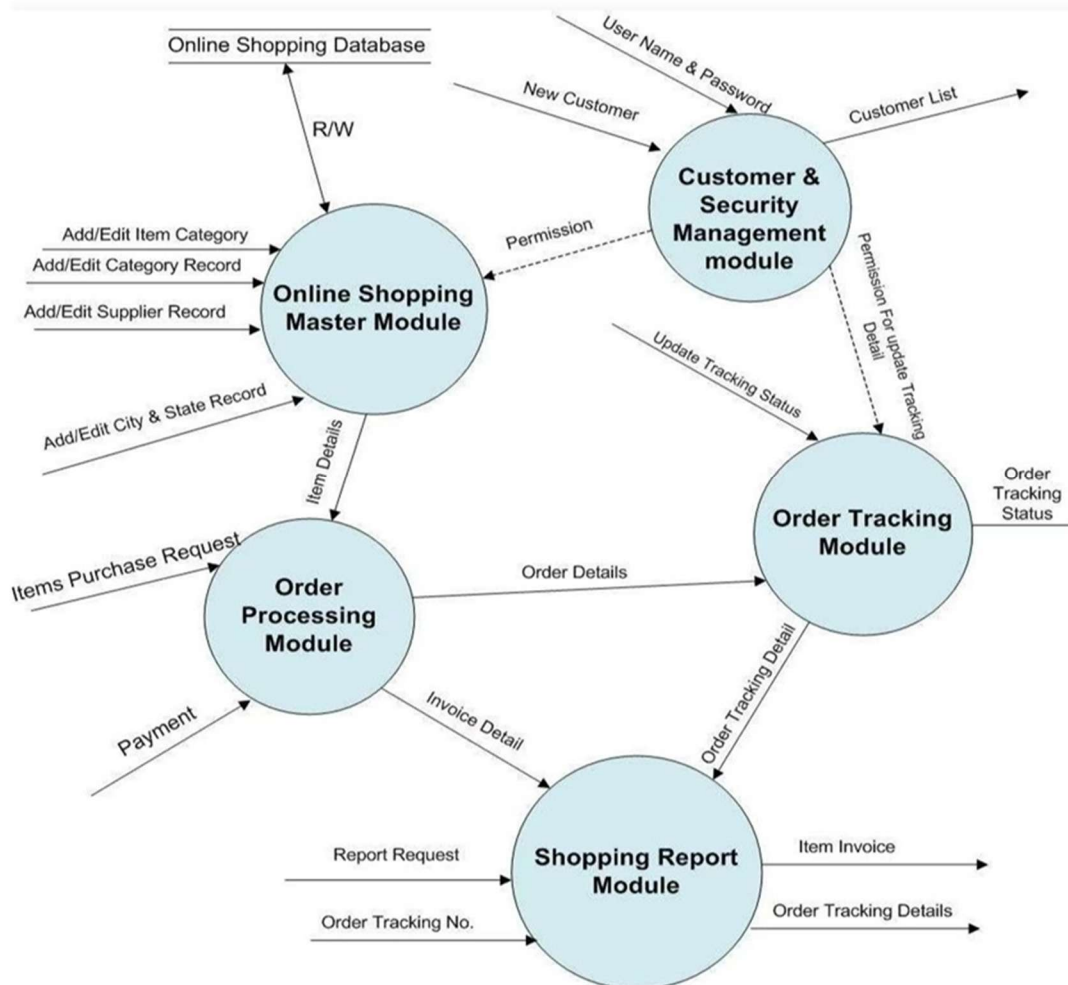
3.1 Context Diagram:

Data flow diagrams can be used to provide a clear representation of any business function. The technique starts with an overall picture of the business and continues by analyzing each of the functional areas of interest. This analysis can be carried out to precisely the level of detail required. The technique exploits a method called top-down expansion to conduct the analysis in a targeted way.

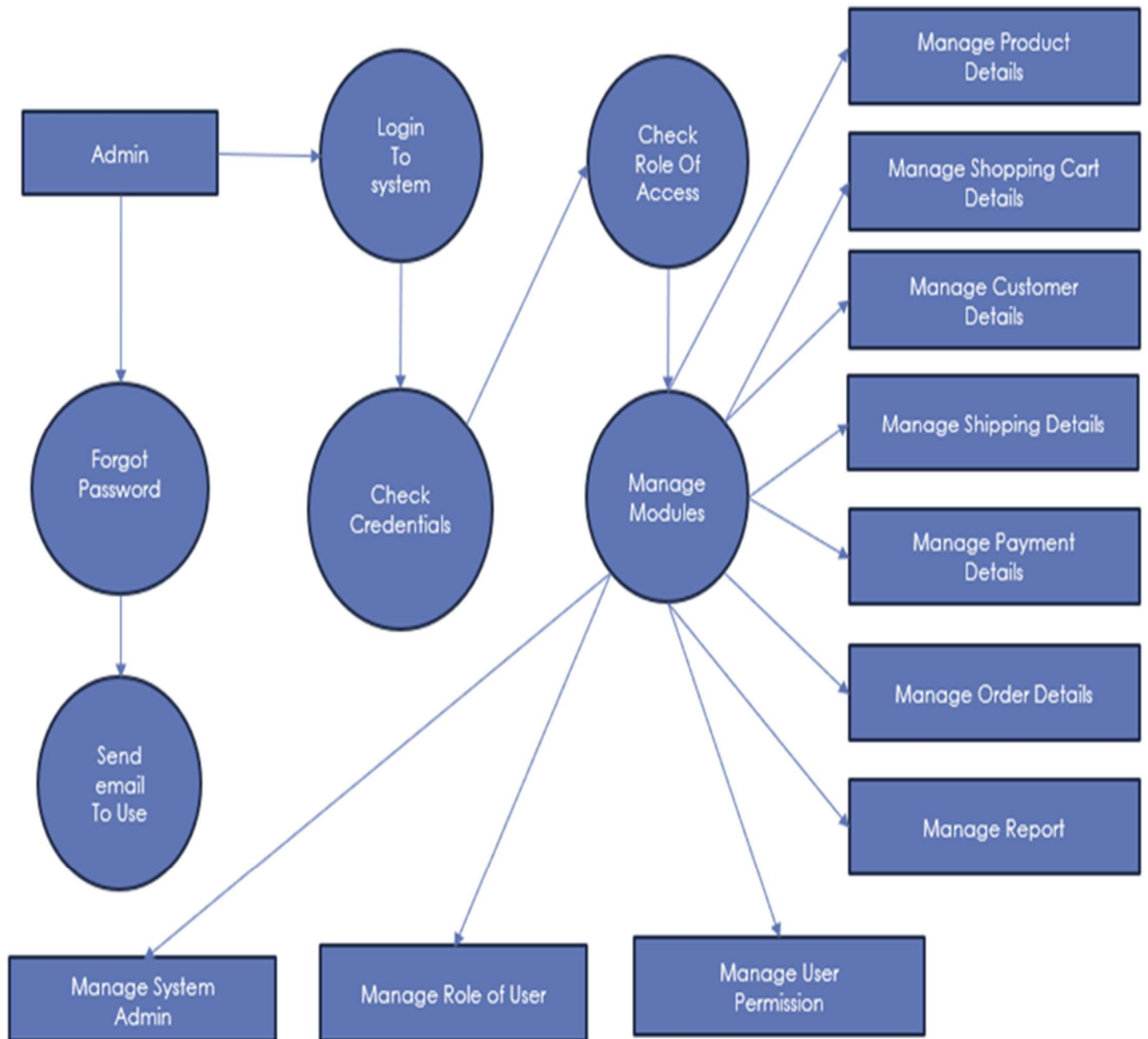
- **Customer** (browsing jewellery, adding to cart, making payment)
- **Admin** (managing products, orders, categories, users)
- **System** (central website handling all operations)
- Possibly **Payment Gateway** (for online payments)



1ST LEVEL DFD



2ND LEVEL DFD

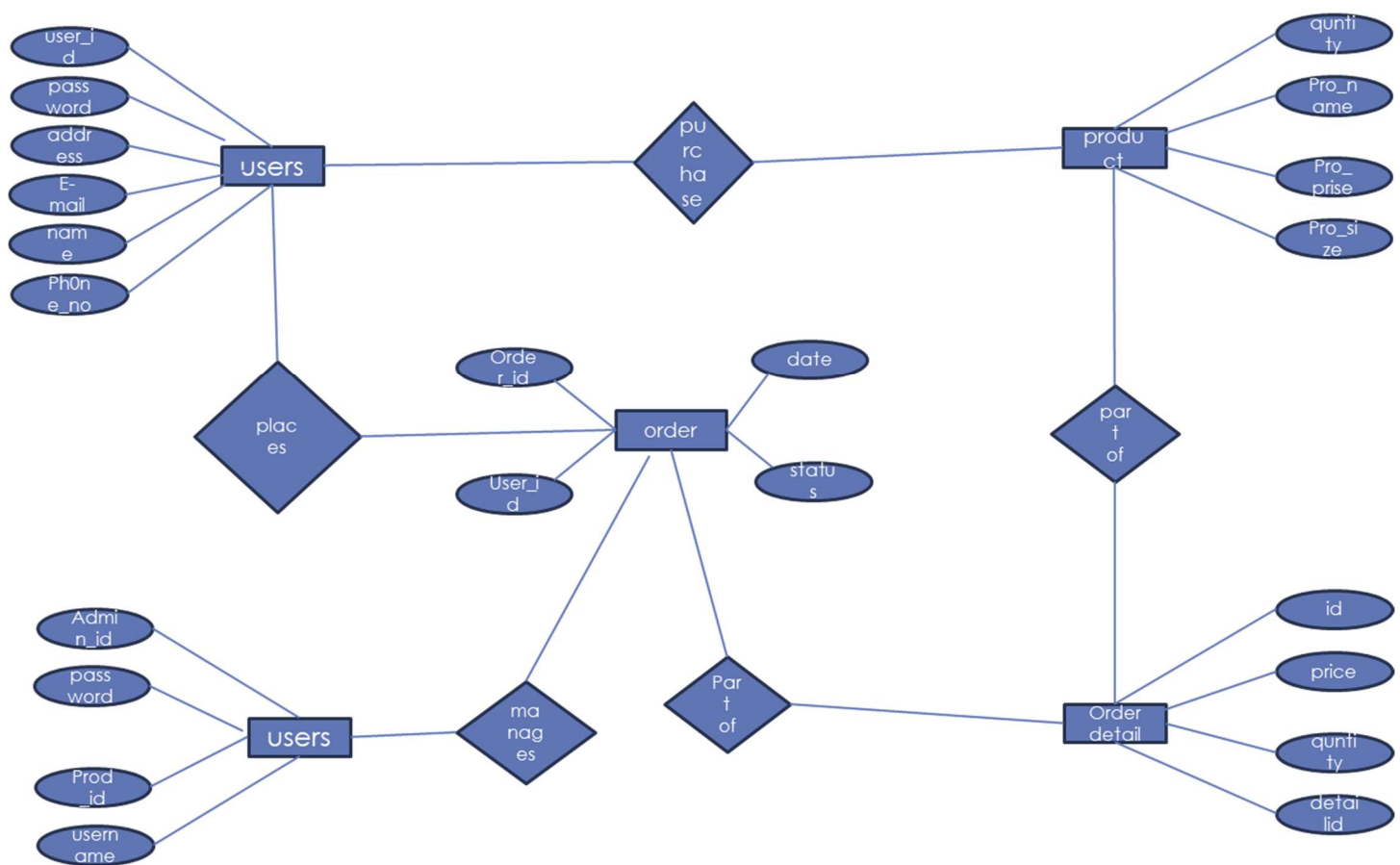


ER-Diagram:

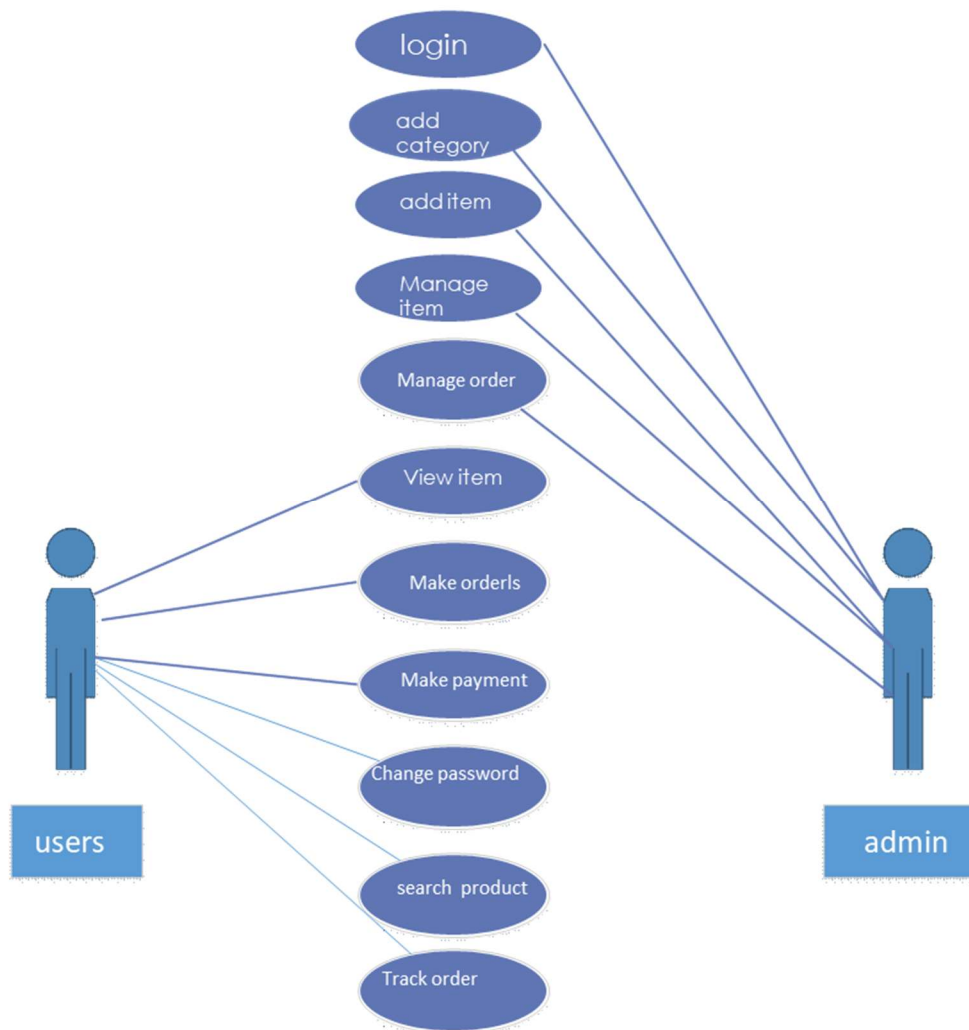
The ER diagram model also called “An Entity Relationship Diagram” is a graphical representation of entities and their relationship to each other, typically used in computing in regard to organization of data within databases or information system.

An entity relationship diagram is a data modeling technique that can help define business process and can be used as foundation for relational database. The shapes we have used in E-R Diagrams are as follows.

E-R Diagram – Database Relationship Chart

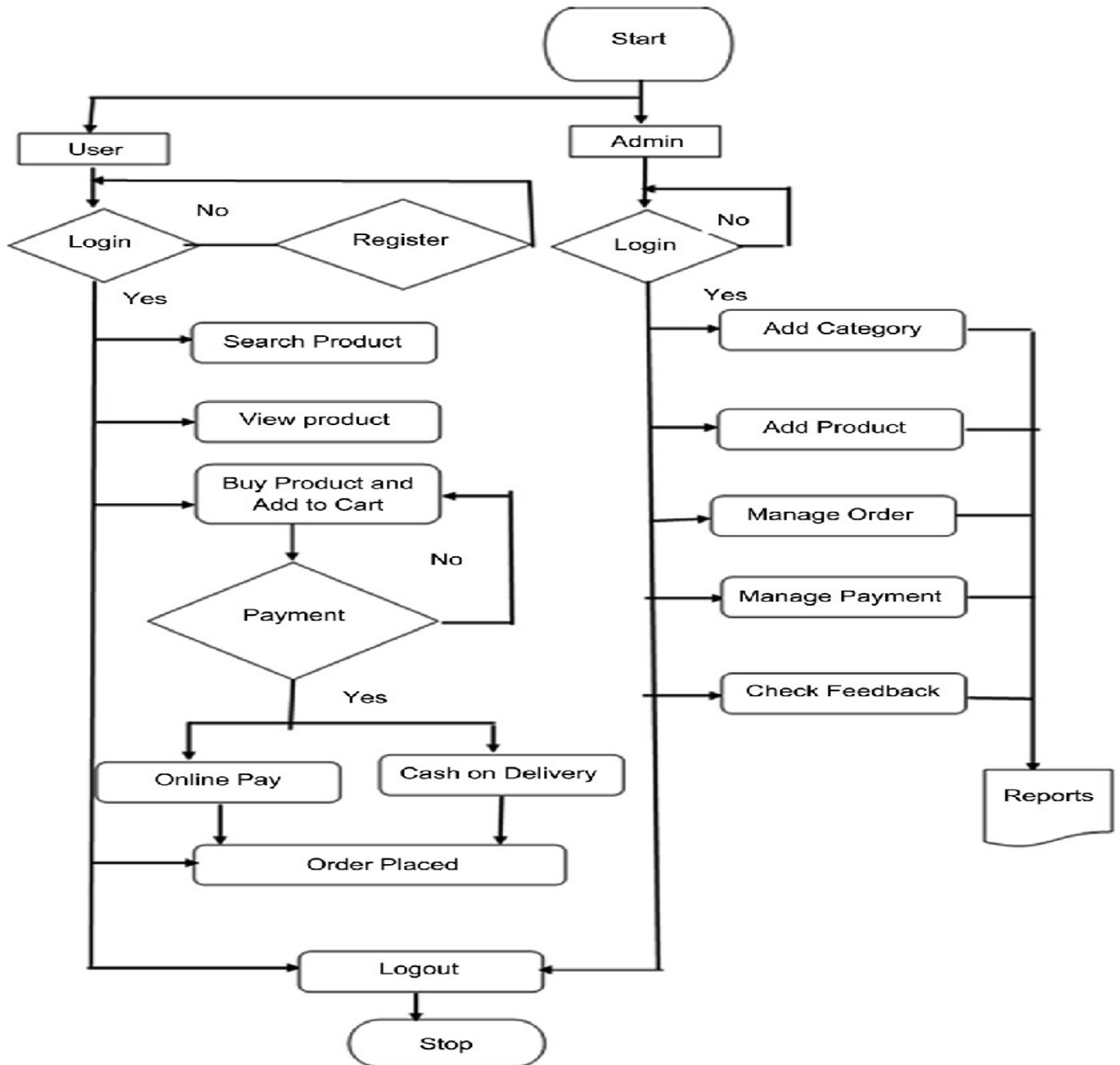


3.4 Use-Cases:



Flow Chart:

PROCESS FLOW DIAGRAM



4. System Design

4.1 Data dictionary

```
import { useEffect, useState } from "react";

function Products() {
  const [data, setData] = useState([]);

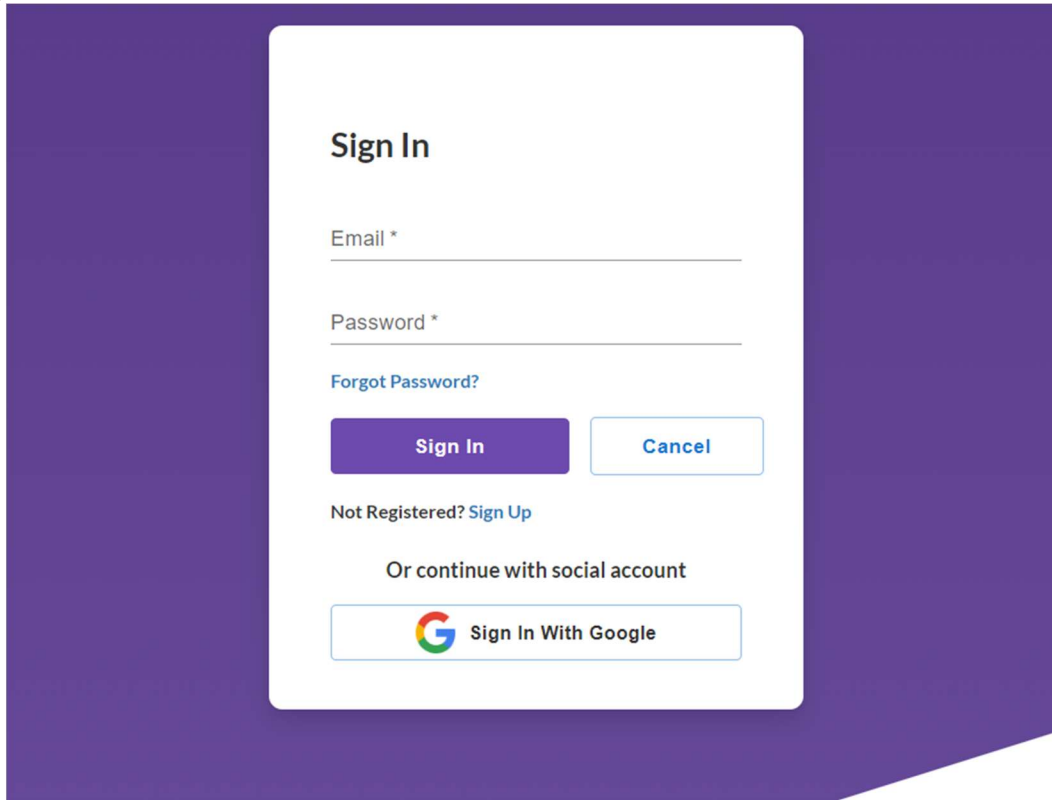
  useEffect(() => {
    fetch("http://localhost:5000/api/products")
      .then(res => res.json())
      .then(result => setData(result))
      .catch(err => console.log(err));
  }, []);

  return (
    <div>
      {data.map((item) => (
        <h3 key={item._id}>{item.name}</h3>
      ))}
    </div>
  );
}

export default Products;
```


4.2 Screenshot of application: At admins

- **Login**



A screenshot of a 'Sign In' form for administrators. The form is centered on a purple background. It includes fields for 'Email *' and 'Password *', a 'Forgot Password?' link, a 'Sign In' button, a 'Cancel' button, a 'Not Registered? Sign Up' link, and a 'Sign In With Google' button.

Sign In

Email *

Password *

[Forgot Password?](#)

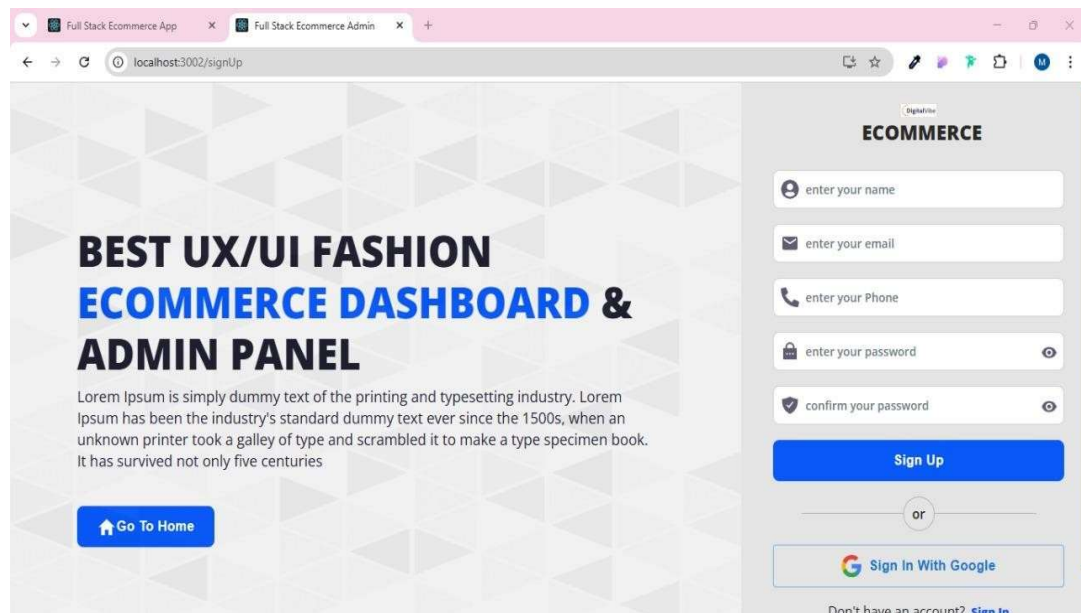
Sign In **Cancel**

Not Registered? [Sign Up](#)

Or continue with social account

 **Sign In With Google**

- **Register**



A screenshot of a web browser showing a 'Full Stack Ecommerce Admin' page. The page has a pink header bar with two tabs: 'Full Stack Ecommerce App' and 'Full Stack Ecommerce Admin'. The browser address bar shows 'localhost:3002/signUp'. The main content area has a light gray background with a geometric pattern. On the left, there is a large text block titled 'BEST UX/UI FASHION ECOMMERCE DASHBOARD & ADMIN PANEL' with a 'Go To Home' button. On the right, there is a 'Sign Up' form for 'ECOMMERCE' with fields for name, email, phone, password, and password confirmation, a 'Sign Up' button, a 'Sign In With Google' button, and a 'Don't have an account? Sign In' link.

Full Stack Ecommerce App x Full Stack Ecommerce Admin x

localhost:3002/signUp

**BEST UX/UI FASHION
ECOMMERCE DASHBOARD &
ADMIN PANEL**

Go To Home

ECOMMERCE

enter your name

enter your email

enter your Phone

enter your password

confirm your password

Sign Up

or

 **Sign In With Google**

Don't have an account? [Sign In](#)

• About

Welcome to DigitalVibe

Your one-stop destination for cutting-edge electronics.

Where Innovation Meets You!

At DigitalVibe, we bring you the latest technology, from smartphones and gaming gear to smart home devices and high-performance laptops. Founded by Rensi Godhani and Heni Suthar, we are committed to offering premium gadgets at unbeatable prices.

- ✓ **Quality You Can Trust** - 100% genuine products
- ✓ **Fast & Reliable Delivery** - Get your orders on time
- ✓ **Secure Shopping** - Safe transactions & encrypted payments
- ✓ **Exclusive Deals** - Best prices & limited-time offers

At DigitalVibe, we are passionate about bringing the latest electronics right to your doorstep. Founded by Rensi Godhani and Heni Suthar, our mission is to offer a seamless and reliable shopping experience with a curated collection of the best gadgets and tech devices.

Our Story

Our journey began with a simple yet powerful idea — to create an online platform that makes purchasing electronics both easy and exciting...

• Product list

Full Stack Ecommerce App

Full Stack Ecommerce Admin

localhost:3002

DigitalVibe

Mahek Chodvadiya
Mahekchodvadiya01@gmail.com

Dashboard

Home Banner Slides

Category

Products

Orders

Home Banners

Home Side Banners

Home Bottom Banners

LOGOUT

Total Users
9

Total Orders
0

Total Products
19

Total Reviews
8

Best Selling Products

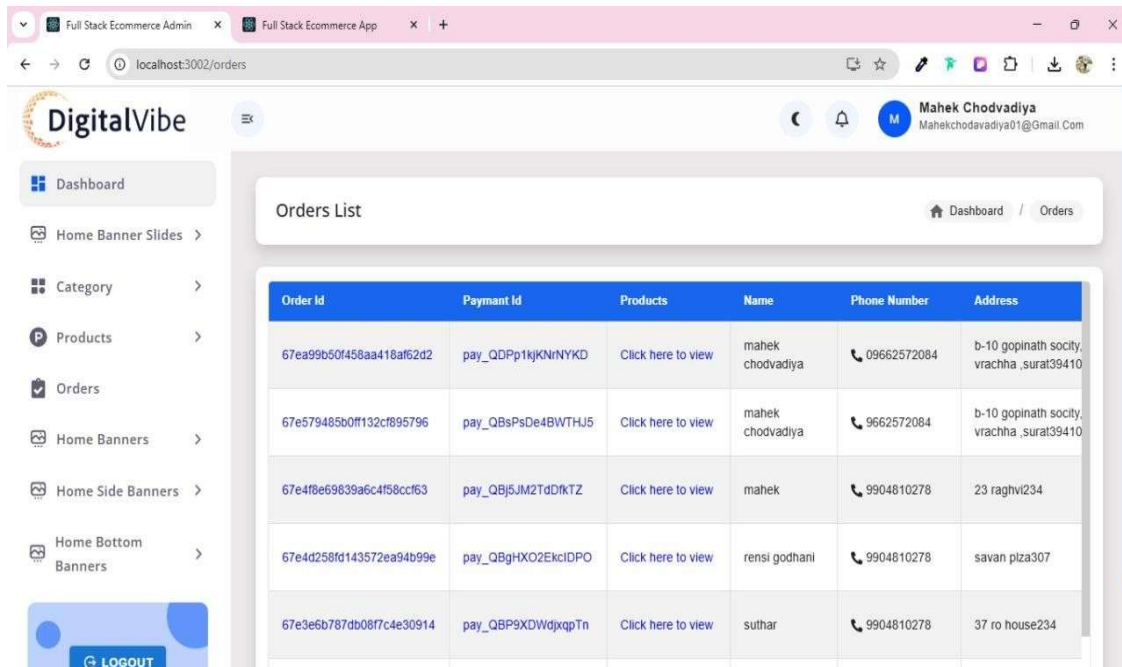
CATEGORY BY
All

Search here...

PRODUCT	CATEGORY	SUB CATEGORY	BRAND	PRICE	RATING	ACTION
 Fridge Croma refrigerators ...	Fridge	croma	croma	Rs-70000 Rs 50000	★★★★★	 
 Fridge A whirlpool is a bod...	Fridge	whirlpool	whirlpool	Rs-70000 Rs 40000	★★★★☆	 
 TV				Rs-100000	★★★★★	 

26 | Page

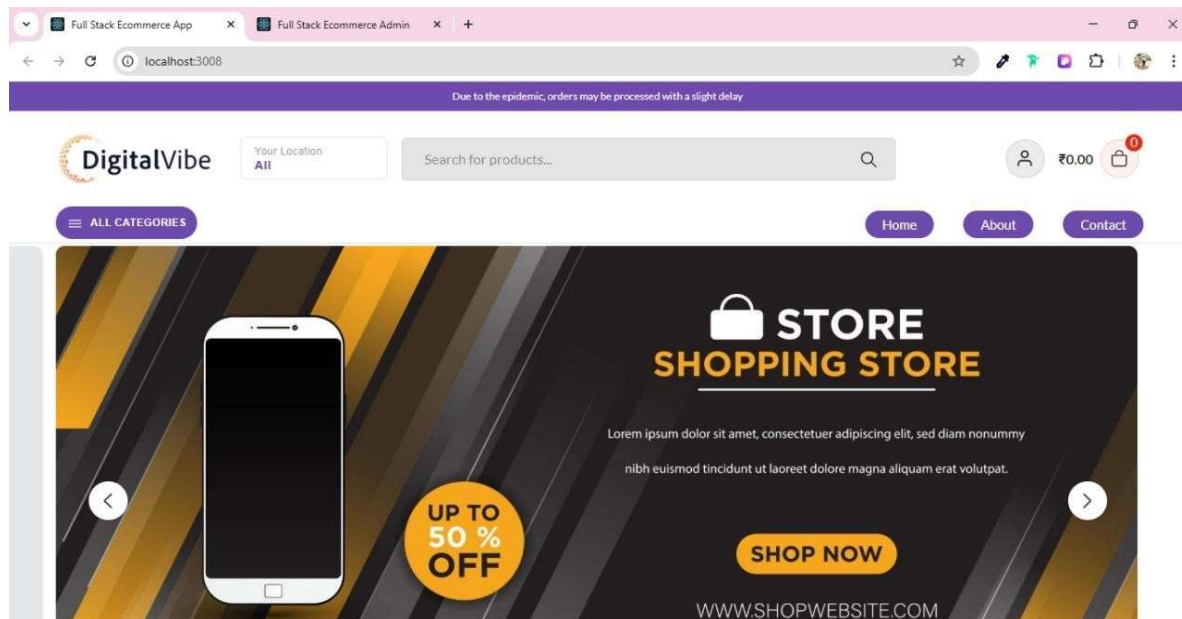
• Order list



Order Id	Payment Id	Products	Name	Phone Number	Address
67ea99b50f458aa418af52d2	pay_QDPp1kjKNrNYKD	Click here to view	mahek chodvadiya	09662572084	b-10 gopinath socity, vrachha, surat39410
67e579485b0ff132cf895796	pay_QBsPsDe4BWTJH5	Click here to view	mahek chodvadiya	9662572084	b-10 gopinath socity, vrachha, surat39410
67e4f8e69839a6c4f58ccf63	pay_QBJ5JM2TdDfKTZ	Click here to view	mahek	9904810278	23 ragthv234
67e4d258fd143572ea94b99e	pay_QBgHXO2EkIDPO	Click here to view	rensi godhani	9904810278	savan plza307
67e3e6b787db087c4e30914	pay_QBP9XDWdjxqpTn	Click here to view	suthar	9904810278	37 ro house234

Client

• Home page



Due to the epidemic, orders may be processed with a slight delay

DigitalVibe Your Location: All Search for products...

₹0.00

ALL CATEGORIES Home About Contact

STORE SHOPPING STORE

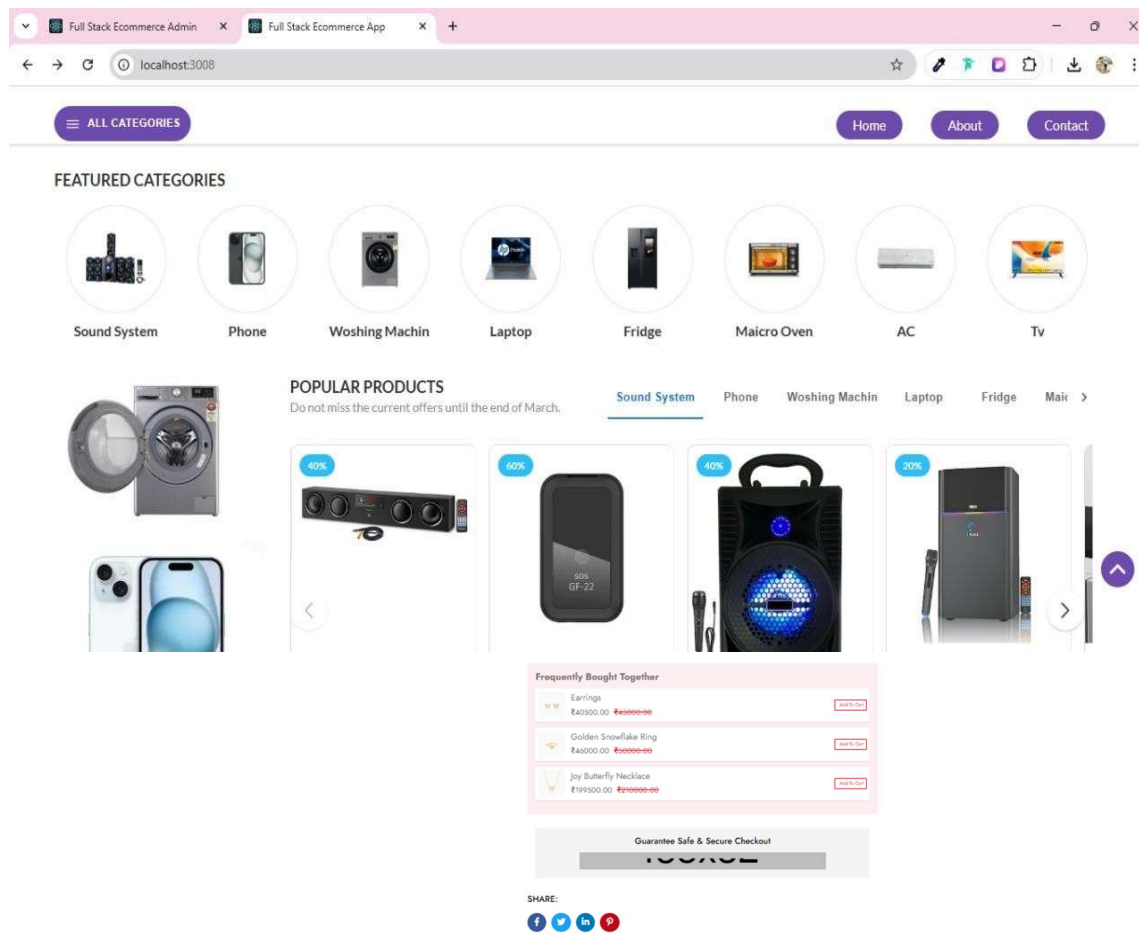
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat.

UP TO 50% OFF

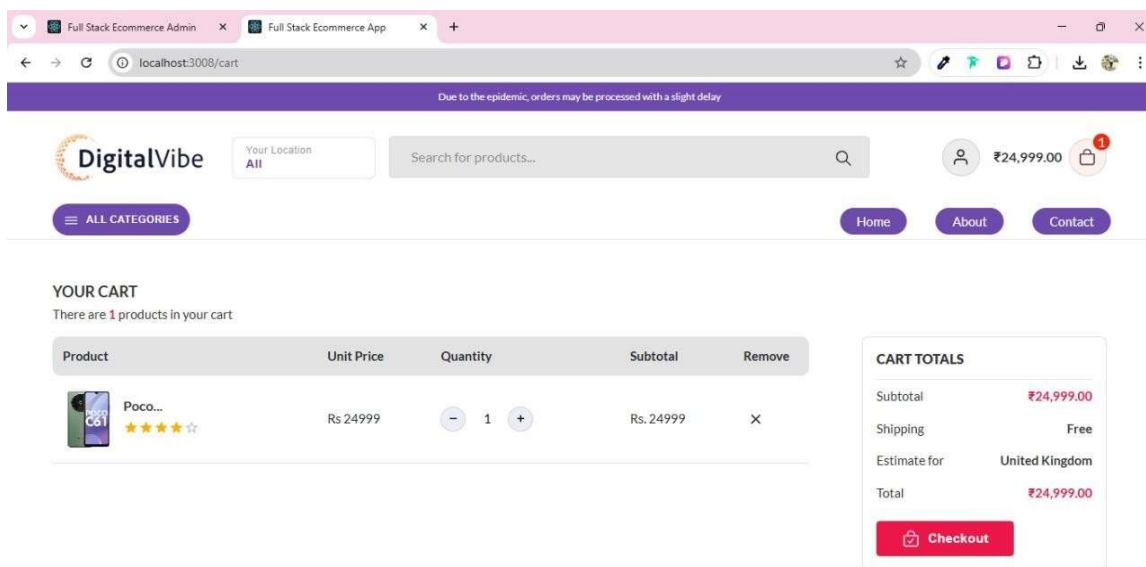
SHOP NOW

WWW.SHOPWEBSITE.COM

• Featured product



• Cart



• Checkout

Due to the epidemic, orders may be processed with a slight delay

DigitalVibe Your Location: All Search for products...

ALL CATEGORIES Home About Contact

BILLING DETAILS

Full Name * mahek chodvadiya Country * India

Street address *
House number and street name b-10 gopinath socity,mota vrachha ,surat

Apartment, suite, unit, etc. (optional) 394101

Town / City *
City surat

State / County *
State gujarat

Postcode / ZIP *
ZIP Code 394101

Phone Number 09662572084 Email Address mahekchodavadiay01@gmail.com

YOUR ORDER

Product	Subtotal
Poco... × 1	₹24,999.00
Subtotal	₹24,999.00

[Buy Now](#)

• Product Detail

Due to the epidemic, orders may be processed with a slight delay

DigitalVibe Your Location: All Search for products...

ALL CATEGORIES Home About Contact

Poco

Brands : Poco ★★★★★ 0 Review

Rs: 30000 **Rs: 24999**

IN STOCK

Rs: Mobile phones have evolved into powerful, multipurpose devices catering to different needs. Choosing the right phone depends on budget, performance, camera quality, display, battery life, and software support. With rapid technological advancements, smartphones continue to enhance communication, entertainment, and productivity, making them an essential part of modern life. 📱

RAM: 2000

Size: 382

Conclusion

In conclusion, the “**Digital Vibe**” project successfully delivers a modern and efficient e-commerce platform for electronic products, simplifying online shopping for customers and management tasks for administrators. The system is developed using HTML, CSS, JavaScript, Bootstrap, and **React.js** for the front end, while **Node.js** and **MongoDB** are used for the back end, ensuring a secure, scalable, and well-structured application.

The platform allows customers to register, browse electronic products, view detailed specifications, add items to the cart or wishlist, and place orders easily through a smooth checkout process. At the same time, the admin panel provides complete control over product management, orders, users, and transactions through proper access control.

Overall, the project demonstrates the practical implementation of full-stack web development in building a real-world e-commerce system. It provides a reliable, user-friendly, and secure solution that meets modern online shopping requirements and can be further enhanced with additional features in the future.

Limitations

- **Limited Payment Options:**

Currently supports basic payment methods; integration with advanced gateways like UPI, wallets, or real-time payment APIs can be enhanced.

- **Scalability:**

The system may require further optimization and cloud integration to efficiently handle a very large number of users and transactions.

- **Customization:**

Users have limited options for personalizing the interface and product display according to their preferences.

- **Security Measures:**

Standard security practices are implemented, but advanced features like data encryption, token-based authentication, and secure APIs can be improved.

- **Mobile Optimization:**

Although the platform is responsive, additional improvements can enhance performance and user experience on mobile devices.

5. Bibliography

- **Book Reference:**

- React.js Documentation – Official React Guide
- Node.js Design Patterns and Development
- Full Stack Web Development with React and Node.js

- **WEB Reference:**

- Official React Website: <https://react.dev/>
- Official Node.js Website: <https://nodejs.org/>
- Official MongoDB Website: <https://www.mongodb.com/>